

https://docs.google.com/spreadsheets/d/10DwonsxZVJQYTz804B1ZqrfFB02eQjBO/edit?usp=drive_link&oid=100190331171335997981&rtpof=true&sd=true

****I have attached a few service line items that are also applicable for positive cases. Tell me how they are generally handled.****

Below is the rule that is as per the Billing guidelines for handling certain cases, Looking at the past cases tell me how it is followed:

****Isolation Care (ICU): Billed to the patient who require isolation. These charges are billed per day in addition to the ICU ward charges.****

****HBsAg/HCV Management-General:** Billed to all surgical patients (except CT Surgery & Cath Lab) when HBsAg/HCV indication sticker (green) is attached to the Case Sheet Cover. Related consumables are billed separately from the pharmacy and are not Included in this amount. No HBsAg/HCV Management charges will be billed for medical management.**

****HBsAg/HCV Management-Cath Lab: Billed to all Cath Lab Procedures when HBsAg/HCV indication sticker (green) is attached to the Case Sheet Cover. Related consumables are billed separately from the pharmacy and are not included in this amount.****

****For HIV +ve cases, the following to be billed according to the LOS:****

****Virology +ve Management Category II. For LOS between 4 to 7 days****

****Virology +ve Management Category IV: For LOS exceeding 10 days****

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****OT Charges for Infective cases HBS Ag , HCV Cases , 50% Extra and for Seropositive cases 100% extra****

The goal is that We understand how to handle positive cases in billing. What is the exact logic we need to follow? And it could differ across cash, Gipsa, non-Gipsa, if that's true

Codex Output

The positive-case logic should be implemented as a separate billing-rule layer driven by verified infection status, clinical context, payer agreement, room/ICU days and LOS. It should not be inferred merely from investigations such as HBsAg testing.

The attached workbook contains seven active codes covering the HBsAg investigation, disposable kit, retropositive charge and three HBsAg/HCV management contexts, with room-category rates. + Cases billing services.xlsx (Sheet1!A1:J8)

What the historical IPs show

There are 124 clean admissions containing a genuine positive-case management charge. All 124 are HBsAg/HCV cases:

Payer	Admissions
Cash	42
Non-GIPSA	42
GIPSA	27
Corporate	13
Total	124

Billing structure:

- 104 procedure admissions.
- 20 medical-management admissions.
- 62 package cases.
- 62 open-bill cases.
- 128 management lines across 124 admissions because four admissions contained two management codes.

Observed codes:

Code	Description	Admissions
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RNS012 3	HBsAg/HCV Management – Non-Heart Surgeries	83
RNS011 6	HBsAg/HCV Management – General	31
RNS012 2	HBsAg/HCV Management – Cath Lab	13
RNS012 1	HBsAg/HCV Management – CT Surgery	1

Every management line had quantity one. These are one-time management charges, not daily charges.

Package handling

Of the 62 package cases:

- 59—95%—showed the HBsAg/HCV management charge as a package exclusion.
- One showed it as included.
- One showed it in both included and excluded evidence.
- One did not have sufficient package classification evidence.

The production default should therefore be:

Positive-case management charges are outside the package unless the applicable package agreement explicitly includes them.

Related pharmacy consumables should also remain separate actuals.

Medical-management discrepancy

The policy says HBsAg/HCV management must not be billed for medical-management admissions.

History does not consistently follow this:

- 19 medical cases were charged RNS0116.
- One medical case was charged RNS0123.
- These cases had no mapped operation or cath-lab procedure.

That is 20 of 124 positive-management admissions. The FC Builder should follow the governed policy and exclude this charge for medical management. The 20 historical cases should be treated as billing exceptions, not estimator precedent.

Duplicate/context overlap

Four admissions had two management codes:

- Two contained both a Cath Lab and non-heart procedure context, so two different charges may be legitimate.
- Two contained both **RNS0116** General and **RNS0123** Non-Heart for effectively the same surgical context and appear duplicative.

The Builder must prevent **RNS0116** and **RNS0123** from being charged together for the same procedure/context.

Final positive-case logic

1. HBsAg/HCV management

Required trigger:

- Verified HBsAg/HCV-positive status or approved green-sticker indication.
- A qualifying surgical or procedural treatment.
- Do not trigger from the **MIC0066** HBsAg investigation alone.

Code selection:

Clinical context	Default code	Quantity
Non-heart surgery	RNS0123	1
CT surgery	RNS0121	1
Cath-lab procedure	RNS0122	1
Medical management	No charge	0

RNS0116 should be retained as a legacy/agreement-specific General code. Historical use is mixed, including inappropriate medical cases. Recommended default:

- Use **RNS0123** for non-heart surgery.

- Use **RNS0116** only when the selected tariff/agreement explicitly specifies it.
- Never add both for the same treatment context.

If one admission genuinely contains separate non-heart and cath-lab procedures, separate applicable management charges may be permitted, with each linked to its procedure/session.

2. HBsAg investigation and consumables

MIC0066 is an investigation, not proof of a positive result.

It appeared in 2,840 clean admissions, including 905 medical-management admissions. It must therefore never trigger positive-case billing by itself.

CAR0146, Disposable Kit HBsAg/HCV, is a separately tariffed item. It had no historical observations in the clean IP service surface.

Rules:

- HBsAg/HCV test: bill as an investigation when ordered.
- Positive-case management charge: bill only after verified positive/sticker status and qualifying procedure.
- Disposable kit and pharmacy consumables: bill separately when used.
- Do not include the consumables automatically in the management charge.
- Do not automatically add **CAR0146** merely because the patient is positive; require cath/cardiology usage evidence.

3. HIV/seropositive management

There are no historical observations of **MSC2816** or the Virology +ve category codes in the clean IP billing surface. This branch must initially be policy-driven.

Use the normalized billable LOS:

Admission type/LOS	Code	Rule
Daycare	HSP5020	Virology +ve Daycare
1–3 days	HSP5021	Category I
4–7 days	HSP5022	Category II

8–10 days HSP502 Category III
3

More than 10 days HSP502 Category IV
4

Rules:

- Daycare takes precedence over the LOS bands.
- The selected category is a flat one-time charge, quantity one.
- Exactly one virology category can be selected.
- Use normalized billable LOS, not simply calendar-date subtraction.

MSC2816 conflict

The attached workbook gives MSC2816 as:

- ₹10,000 for General/Twin/Single/ICCU.
- ₹5,000 for Daycare. + Cases billing services.xlsx (Sheet1!A4:J4)

However, the current clean tariff views contain ₹10 placeholder-like rates for the same code.

Therefore:

- Do not activate MSC2816 using the current database rate.
- Ingest the attached rates as a new source only after assigning payer/tariff scope and effective date.
- Treat MSC2816 and HSP5020–HSP5024 as mutually exclusive charging schemes unless an agreement explicitly says both apply.
- The preferred general HIV logic is the LOS-based HSP5020–HSP5024 structure.
- MSC2816 should remain a legacy or payer-specific alternative.

Isolation handling

Generic isolation

When `requires_isolation = true`:

- Charge normal room/ICU occupancy as usual.
- Add Isolation Care Room for every eligible isolation-room day.
- Add Isolation Care ICU for every eligible ICU-isolation day.
- Calculate quantities from the day-level room ledger.
- Never apply both room and ICU isolation care to the same patient-day.

Formula:

$$\text{room_isolation_amount} = \text{isolation_room_days} \times \text{applicable_room_isolation_rate}$$
$$\text{icu_isolation_amount} = \text{isolation_icu_days} \times \text{applicable_icu_isolation_rate}$$

These charges are additive to normal room and ICU charges.

H1N1

For verified H1N1-positive cases:

- Add the applicable isolation-room/ICU occupancy charges.
- Add the relevant isolation-care charge for each isolation day.
- A test being ordered is not sufficient; the positive result or approved clinical isolation status is required.

Current data gap

There are no isolation-care billing lines in the clean historical IP service data.

The current tariff views contain:

- **RNS0101**: Isolation Care (Room).
- **ROM5058**: Isolation Care.

But no clearly governed ICU-specific isolation-care code has been identified. The Builder should not invent one. ICU isolation must remain **CONTEXT_REQUIRED** until its code and payer/room rates are supplied or the hospital confirms that **RNS0101** is also used for ICU with an ICCU rate.

Non-GIPSA OT surcharge

The MOU rule says:

- HBsAg/HCV infective cases: 50% extra OT charge.
- HIV/seropositive cases: 100% extra OT charge.

Recommended calculation:

$$\text{infective_ot_surcharge} = \text{eligible_ot_base} \times 50\%$$
$$\text{seropositive_ot_surcharge} =$$

$\text{eligible_ot_base} \times 100\%$

The surcharge should be stored as a separate calculation line. Do not overwrite the underlying OT tariff rate.

By default, **eligible_ot_base** should include only the actual OT room/time charges in the **OT CHARGES** service group. It should not automatically include:

- OT instruments.
- Anaesthesia PF.
- Surgeon PF.
- Cath-lab charges.
- Pharmacy.
- Implants.
- Recovery charges.

Those can be added only if the applicable MOU explicitly includes them.

If a patient has multiple positive classifications, use the highest applicable surcharge rather than adding 50% and 100%, unless the MOU specifically permits cumulative surcharges.

Historical validation

The historical service lines do not show this uplift:

Payer	Matched OT lines	Median positive/control rate
Cash	38	1.03×
Corporate	18	1.00×
GIPSA	48	1.00×
Non-GIPSA	63	1.00×

For Non-GIPSA:

- 90.5% of matched OT lines were close to the normal 1.00× rate.
- None were close to 1.50×.

Therefore the 50% OT uplift is not present in the historical LAN/service surface. It may be a final-insurance claim adjustment or may not have been consistently applied.

Implementation position:

- Cash: no automatic infective OT surcharge unless the Cash policy supplies one.
- GIPSA: no automatic surcharge unless the TR290/GIPSA agreement supplies one.
- Non-GIPSA: apply only when the patient's organization/MOU explicitly contains the 50%/100% rule.
- Corporate: organization-specific.
- Mark these as final-insurance adjustments, not LAN defaults, unless the agreement says otherwise.

Payer-specific decision table

Rule	Cash	GIPSA	Non-GIPSA	Corporate
Clinical positive-case eligibility	Same clinical trigger	Same clinical trigger	Same clinical trigger	Same clinical trigger
Rate source	Cash tariff	TR290/GIPSA agreement	Organization/MOU tariff	Corporate agreement
HBsAg/HCV package handling	Outside package unless included	Usually exclusion; verify GIPSA package	Agreement/package-specific	Agreement-specific
Infective OT +50%	No default	No default	Apply only when MOU specifies	Agreement-specific
Seropositive OT +100%	No default	No default	Apply only when MOU specifies	Agreement-specific
Virology LOS category	Cash tariff	TR290 rate	MOU tariff	Corporate tariff
Isolation care	Applicable tariff per day	TR290/policy	MOU/policy	Agreement-specific

Required FC input fields

Add a positive-case section to the FC flow:

positive_status:

NONE

HBSAG

HCV

HIV_SEROPOSITIVE

H1N1
OTHER_INFECTIVE

confirmation_source:
GREEN_STICKER
POSITIVE_LAB_RESULT
CLINICAL_DIAGNOSIS
MANUAL_APPROVED

requires_isolation: true/false
projected_isolation_room_days
projected_isolation_icu_days
surgery_context:
NONE
NON_HEART_SURGERY
CT_SURGERY
CATH_LAB
payer_agreement_id

The Builder should show these questions only when relevant, but a positive-case flag should be available in the initial flow because it changes package exclusions, management charges and potentially OT billing.

Rate-selection rule

The values in the attached workbook should not be hard-coded universally. Historical exact matches to those rates were only approximately:

- RNS0116: 36%.
- RNS0122: 11%.
- RNS0123: 41%.

The remaining cases used different payer, tariff, room or effective-date rates.

Always resolve rates using:

service code
+ payer tariff
+ organization/MOU
+ room category
+ effective date

Production-ready rule status

Ready to implement:

- HBsAg/HCV management restricted to qualifying procedures.
- Context-specific [RNS0121/RNS0122/RNS0123](#) selection.
- Quantity one for management charges.
- No HBsAg/HCV management charge for medical management.
- Positive management and related consumables outside packages unless explicitly included.
- HIV LOS-category selection.
- Isolation care using day-level quantities.
- Payer/MOU-based OT surcharge calculation.
- A positive test order alone does not trigger positive-case billing.

Blocked pending master/agreement clarification:

- ICU isolation-care service code and rates.
- Effective date and payer scope of the attached workbook.
- [MSC2816](#) ₹10 versus ₹10,000 master conflict.
- Whether [RNS0116](#) remains valid for any current tariff.
- Exact Non-GIPSA organizations/MOUs using 50%/100% OT uplifts.
- Whether package cases calculate the OT surcharge on a separately reconstructed OT tariff base.
- Whether [MSC2816](#) is an alternative to or additional to the Virology +ve LOS categories.

No database or FC Builder logic was changed during this review.

Executive summary

This defines a **new Positive Case Billing Rule Engine** for the FC Builder. The logic should **not** be inferred from investigations alone—it should be driven by **confirmed infection status**, **clinical context**, **payer agreement**, **package rules**, and **LOS/room usage**.

Key implementation decisions

- **HBsAg/HCV management charges**
 - Apply **only** when the patient is a **verified positive case** and undergoes a qualifying procedure.
 - Do **not** trigger from the HBsAg investigation ([MIC0066](#)) alone.
 - Use context-specific codes:

- Non-heart surgery → RNS0123
 - CT surgery → RNS0121
 - Cath Lab → RNS0122
 - Medical management → **No charge**
- Quantity is always **1** (one-time charge).
- **Packages**
 - Positive-case management charges should be treated as **outside the package by default**, unless the payer/package agreement explicitly includes them.
 - Related consumables remain separate actuals.
- **Medical management**
 - Even though historical data contains 20 medical admissions that were billed, these are treated as **historical billing exceptions**.
 - FC Builder should **never estimate** these charges for medical-management admissions.
- **Duplicate prevention**
 - Prevent RNS0116 and RNS0123 from being billed together for the same surgical context.
 - Allow multiple management charges only when they correspond to genuinely different procedures (e.g., Cath Lab + surgery).
- **Investigations vs positive status**
 - MIC0066 (HBsAg test) is only an investigation.
 - A test order **does not imply** the patient is positive.
- **HIV / Seropositive**
 - Use LOS-based category codes (HSP5020–HSP5024).
 - Select exactly **one** category.
 - MSC2816 remains blocked until tariff conflicts are resolved.
- **Isolation**
 - Isolation charges are **daily** and based on actual isolation room/ICU days.
 - Add isolation charges **in addition to** normal room charges.
 - ICU isolation remains blocked until the hospital confirms the correct service code.
- **OT surcharge**
 - MOU rules:
 - HBsAg/HCV → 50% OT surcharge.
 - HIV → 100% OT surcharge.
 - Apply **only** where the specific payer/MOU requires it.
 - Store the surcharge as a separate calculation line.
 - Historical billing does **not** consistently show these uplifts, so they should **not** be default behavior.
- **Rate resolution**
 - Never hard-code workbook rates.
 - Always resolve rates using:
 - Service code
 - Payer tariff

- Organization/MOU
- Room category
- Effective date

New FC Builder inputs

Add a Positive Case section containing:

- Positive status (HBsAg/HCV/HIV/H1N1/etc.)
- Confirmation source (lab, green sticker, clinical diagnosis, manual approval)
- Isolation requirement
- Isolation room/ICU days
- Surgical context
- Payer agreement

Show these fields only when relevant.

Implementation status

Ready

- Positive-case eligibility logic
- Context-based code selection
- Package exclusion logic
- One-time management charges
- HIV LOS categorization
- Isolation day calculations
- MOU-driven OT surcharge framework
- Investigation ≠ positive case

Blocked

- ICU isolation service code/rates
- Effective dates and payer scope of the supplied workbook
- **MSC2816** tariff conflict (₹10 vs ₹10,000)
- Current validity of **RNS0116**
- Which Non-GIPSA organizations actually use OT uplift
- Package OT surcharge calculation methodology
- Whether **MSC2816** replaces or coexists with the LOS-based HIV categories

Overall, this moves positive-case billing from **simple service-code inference** to a **governed billing-rule layer** that is driven by verified clinical status, treatment context, payer agreements, and package policies.